

Article

A New Design of MIMO Antenna with Dual-Band/Dual-Polarized Modified PIFAs for Future Handheld Devices

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Abstract

This paper introduces a compact sub-6 GHz multiple-input multiple-output (MIMO) antenna array developed for 5G smartphone applications. The design employs eight planar inverted-F antenna (PIFA) elements arranged to realize dual-band and dual-polarized operation. The antenna achieves impedance bandwidths of 3.3–3.7 GHz (11.4%) and 5.3–5.8 GHz (10%), covering key sub-6 GHz fifth-generation (5G) bands. To enhance diversity performance, the elements are distributed along the edges of the smartphone mainboard, enabling excitation of orthogonal polarization modes while maintaining an overall board size of 75 mm × 150 mm on an FR4 substrate. Even without the use of dedicated decoupling structures, the closely spaced antenna elements exhibit satisfactory isolation levels, varying between −12 dB and −22 dB across the operating bands. The antenna array achieves wide impedance bandwidths of approximately 400 MHz at 3.5 GHz and more than 500 MHz at 5.5 GHz, supporting high data-rate communication. In addition, the proposed system demonstrates very low correlation and active reflection, with envelope correlation coefficient (ECC) values below 0.002 and total active reflection coefficient (TARC) levels better than −20 dB. User interaction effects are also investigated, and the results confirm acceptable SAR levels and stable radiation behavior in the presence of the human body. Owing to its planar, dual-band/dual-polarization capability and compliance with safety requirements, the proposed antenna represents a promising practical solution for contemporary 5G handheld devices and future multi-band mobile platforms.

Keywords: dual-band antennas; PIFA; polarization diversity; future smartphones



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1. Introduction

The increasing demand for high data rates, reliable connectivity, and support for emerging wireless services has driven continuous advancements in cellular communication systems. To overcome the performance limitations of fourth-generation (4G) networks, fifth-generation (5G) wireless technology has been developed, offering enhanced capacity, improved reliability, and support for diverse application scenarios [1,2]. In particular, sub-6 GHz frequency bands continue to play a vital role in 5G deployments due to their balanced trade-off between coverage and throughput. 5G systems are expected to support a wide range of services, such as enhanced mobile broadband, which require efficient utilization of the available spectrum. To meet these requirements, modern mobile terminals increasingly rely on multi-antenna solutions to improve link robustness and spectral

efficiency. As a result, multiple-input multiple-output (MIMO) techniques have become an integral part of cellular systems [3].

The adoption of MIMO in wireless terminals enables performance enhancement by exploiting multiple propagation paths, leading to improved data throughput and more reliable communication links. This concept has been successfully implemented in 4G Long-Term Evolution (LTE) systems and has gained even greater importance in 5G networks, where higher performance targets necessitate more advanced antenna configurations [4]. However, implementing multiple antennas in compact devices such as smartphones introduces several design challenges. In practical smartphone platforms, antenna elements must be placed within a limited physical area, often resulting in strong mutual coupling and increased signal correlation. To mitigate these effects, diversity techniques—such as spatial and polarization diversity—are commonly employed, allowing the transmission and reception of uncorrelated signals and improving link stability under multipath fading conditions [5]. These techniques are particularly important for maintaining reliable performance in realistic user environments. Furthermore, modern 5G smartphones are required to operate over multiple frequency bands to support different network deployments and regulatory requirements. Dual- or multi-band antenna operation, combined with compact size, planar geometry, and ease of integration, has therefore become a key design objective for next-generation mobile terminals [6]. These considerations motivate the development of antenna solutions that can simultaneously achieve compactness, multi-band operation, and robust MIMO performance. For MIMO antennas, this feature ensures robust performance in diverse operating environments by efficiently supporting multiple frequency bands. This capability not only enhances spectral efficiency but also enables simultaneous operation across sub-6 GHz frequencies, ensuring high data rates, low latency, and reliable connectivity [7,8]. In addition, MIMO systems benefit from diversity techniques, including spatial, pattern, and polarization diversity. Among these, polarization diversity plays a vital role in improving signal quality and reducing correlation between antenna elements, particularly in compact devices such as smartphones. By employing dual-polarized antennas, orthogonal polarization states can transmit and receive signals simultaneously, effectively mitigating polarization mismatch losses and enhancing link reliability in multipath-rich environments [9].

In recent years, numerous designs have been reported for sub-6 GHz cellular applications [10–26]. However, many existing antennas suffer from limitations such as narrow impedance bandwidth, bulky or uniplanar structures that occupy significant PCB area, and increased system complexity. Moreover, most reported designs support only single-band operation and employ single-polarization configurations, which limit diversity performance and reduce MIMO efficiency in multipath environments. These challenges highlight the need for compact MIMO antennas that offer dual-band and dual-polarized operation with wide bandwidth and high isolation, making them suitable for next-generation 5G smartphones. To address them, a new MIMO antenna is introduced, featuring dual-resonance radiators with a compact PIFA configuration, providing efficient dual-band/dual-polarized coverage. The design is optimized for advanced 5G applications, offering enhanced performance across various frequency bands while maintaining a compact form factor. A straightforward discrete-feeding method has been employed in this design, offering simplicity and ease of integration with other smartphone circuit components. Additionally, its minimal complexity supports cost-effective manufacturing and enhances the overall reliability of the device [27–29]. The suggested antenna structure is realized on an FR-4 material. Each element includes a modified planar inverted-F antenna with an additional strip. Despite lacking dedicated decoupling structures, the adjacent elements within the array demonstrate high isolation. These compact antenna elements

efficiently operate at frequencies of 3.5 and 5.5 GHz, delivering full radiation/polarization coverage. This design ensures comprehensive connectivity and optimal performance across the frequency bands, making it suitable for wireless systems. The properties and performance of the element, as well as the MIMO design, are meticulously evaluated using CST software [30]. Through rigorous simulations, the proposed antenna's capabilities and efficacy are thoroughly investigated, laying the groundwork for an efficient wireless communication solution for 5G smartphones.

2. Critical Properties of the Design

Figure 1 illustrates the configuration of the MIMO antenna array developed for smartphone applications. The antenna system adopts a multi-fed planar architecture with a compact footprint of $75 \times 150 \text{ mm}^2$, making it suitable for integration within a standard smartphone mainboard and the corresponding design parameters are summarized in Table 1. The antenna is fabricated on an FR4 substrate, which is commonly used in commercial smartphone printed circuit boards due to its low cost and compatibility with PCB manufacturing processes. The design consists of eight modified PIFA elements, arranged as closely spaced antenna pairs and distributed along different edges of the printed circuit board. The selected antenna configuration offers several performance-related advantages. The modified PIFA elements contribute to improved impedance matching and frequency response, while their symmetrical arrangement enables effective polarization diversity. In addition, positioning the antenna elements in different regions of the mainboard helps achieve more uniform radiation coverage across the device [31].

Table 1. Values of the design parameters.

Param.	mm	Param.	mm	Param.	mm
W	150	L_2	5	W_5	10
L	75	W_3	11.5	W_6	2
W_1	92.5	$L_3 = L_5 = L_8$	1.25	L_6	5
L_1	17.5	W_4	6.8	W_7	2.5
W_2	23.75	L_4	3.75	L_7	0.25

Figure 2 shows the configurations of the PIFA antenna pairs with and without the additional strip. The reference configuration without the strip supports a single resonant mode and features a compact and symmetric structure suitable for smartphone PCB integration. By introducing the additional strip, the effective electrical length of the radiator is modified, enabling excitation of an additional resonant mode without increasing the antenna footprint or structural complexity [32]. The simulated S-parameter results corresponding to the antenna configurations are illustrated in Figure 3. The antenna without the strip exhibits a single resonance centered at 3.5 GHz, confirming single-band operation. In contrast, the modified design with the added strip excites a second resonance around 5.5 GHz, resulting in dual-band operation at 3.5/5.5 GHz. Although the impedance matching around the first resonance is slightly modified after introducing the strip, the overall matching remains acceptable, and the strip enables the desired dual-band operation. Good impedance matching is achieved in both bands, while low mutual coupling between antenna elements indicates effective MIMO performance.

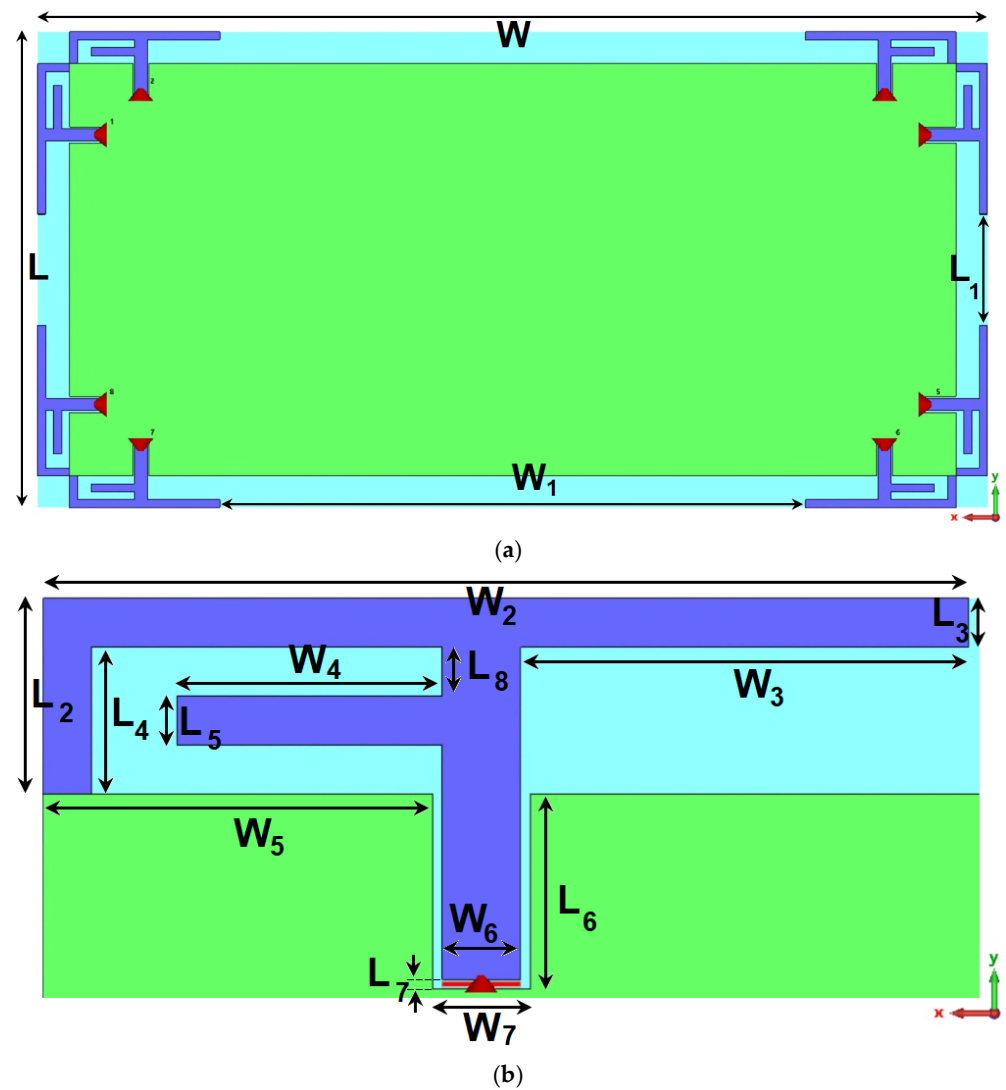


Figure 1. Schematics of (a) the suggested MIMO and (b) the design details of its element.

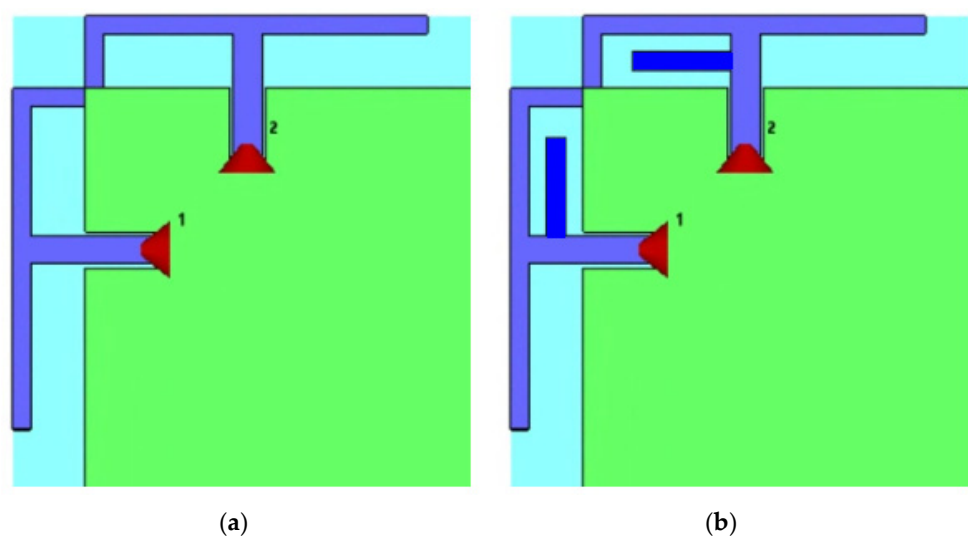


Figure 2. Antenna configurations (a) without and (b) with the strip.

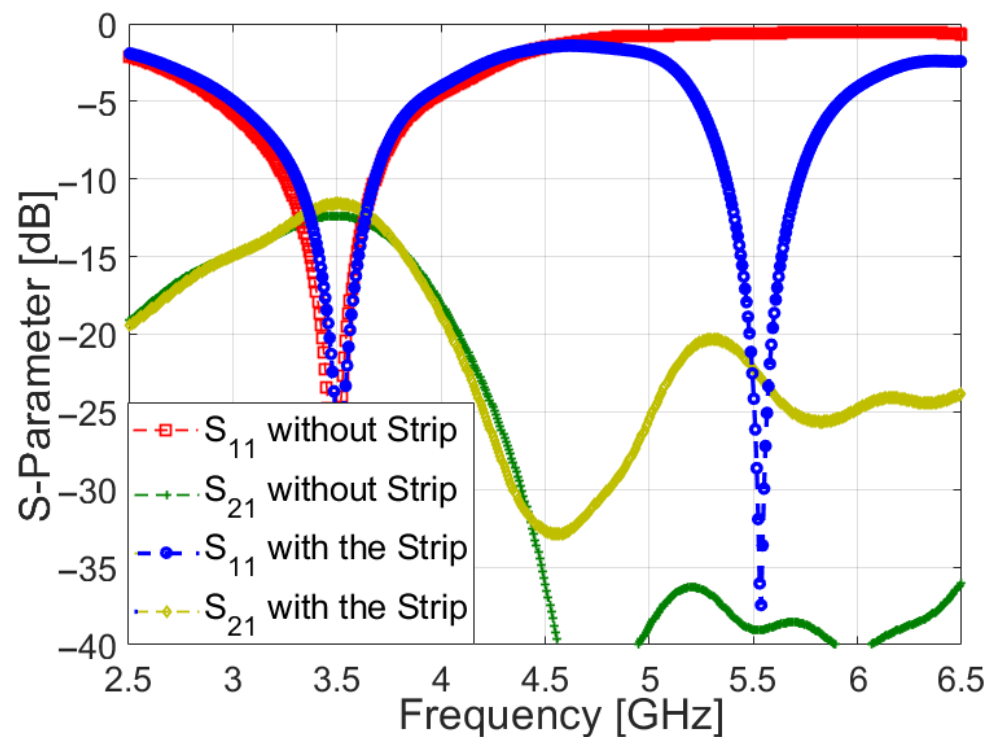


Figure 3. S-parameter results of the designs shown in Figure 2.

To better understand the dual-resonance and dual-band characteristics of the antenna resonators, Figure 4 illustrates the surface current distributions of the proposed dual-polarized PIFA structure. As shown in Figure 4a, in the absence of the strip, the current is mainly concentrated along the PIFA radiator at the resonance frequency of 3.5 GHz, corresponding to single-band operation. When the strip is introduced, the current distribution changes, as shown in Figure 4b, where the strip modifies the current path at 3.5 GHz. Furthermore, as illustrated in Figure 4c, a strong current distribution appears near the strip region at 5.5 GHz, confirming that the strip enables the excitation of an additional resonant mode and results in the observed dual-band behavior [33,34].

Figure 5 presents the simulated S-parameters of the proposed antenna array. The results demonstrate a multi-resonant behavior, with impedance bandwidths covering 3.3–3.7 GHz and 5.3–5.8 GHz for a -10 dB reflection coefficient. When a -6 dB matching criterion is considered, the operating bandwidth is significantly expanded, exceeding 800 MHz in each band, which is well suited for practical 5G smartphone applications. As shown, the reflection coefficients (S_{11}) are notably low, ranging from approximately -25 dB to -35 dB across the operating bands, indicating excellent impedance matching. In addition, the mutual coupling between antenna elements is effectively suppressed, with isolation levels (S_{nm}) better than -11 dB. Notably, this level of isolation is achieved without employing any additional decoupling structures, despite the fact that the PIFA pairs are closely arranged and effectively share adjacent edges with negligible spacing. This inherent isolation capability significantly reduces inter-element interference and enhances overall MIMO performance and signal integrity.

Figure 6 illustrates the three-dimensional radiation patterns of the proposed multi-antenna system, along with the corresponding gain values at 3.5 GHz and 5.5 GHz. The results indicate that the eight-antenna array generates radiation with multiple polarization orientations, confirming effective polarization diversity across the operating bands. Furthermore, the achieved gain levels are sufficient to provide radiation coverage over different regions of the smartphone PCB, ensuring reliable signal transmission in various directions.

This radiation behavior demonstrates that the proposed antenna array is robust with respect to different user-hand holding positions, which is a critical requirement for practical 5G handheld devices. Such robustness helps maintain consistent and stable connectivity under realistic usage scenarios, thereby enhancing overall system performance [35,36].

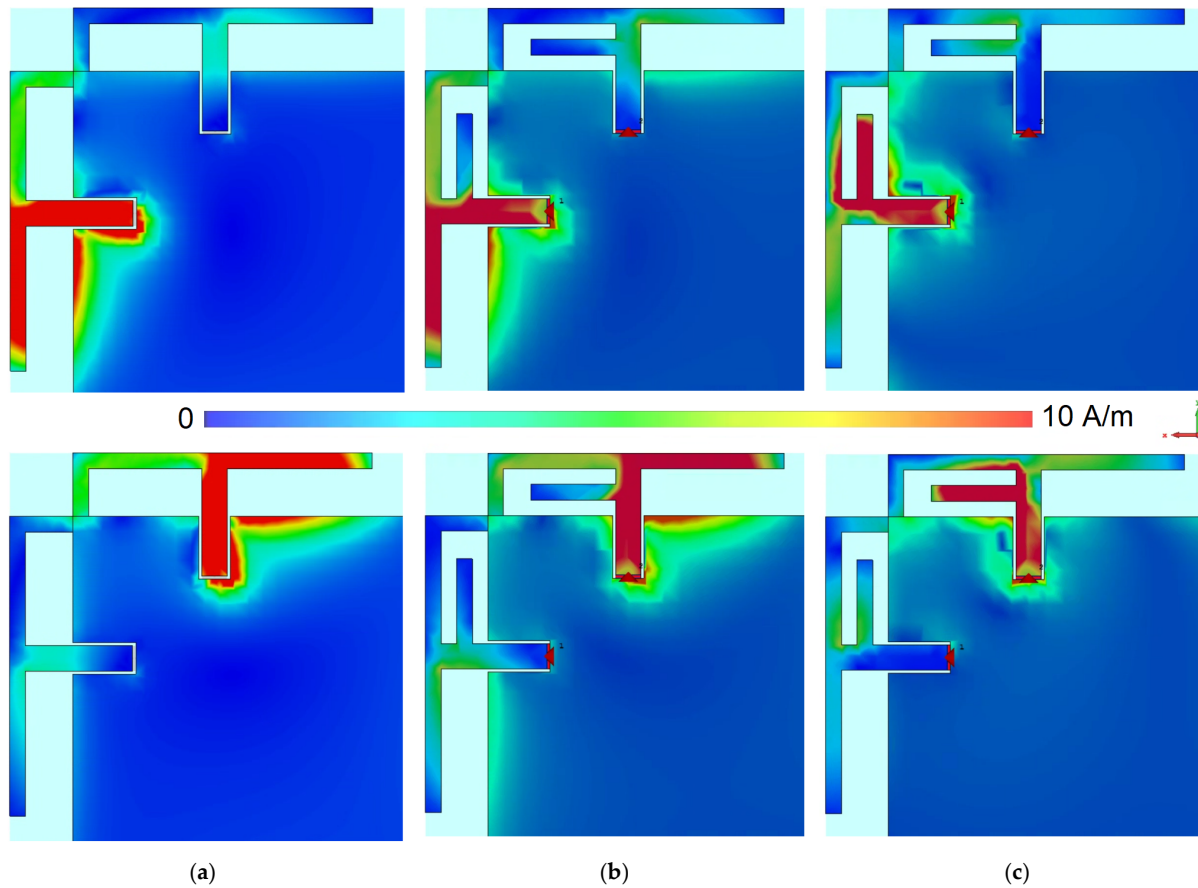


Figure 4. Surface current distributions for the dual-polarized PIFA resonators: (a) without the strip at 3.5 GHz, (b) with the strip at 3.5 GHz, and (c) with the strip at 5.5 GHz.

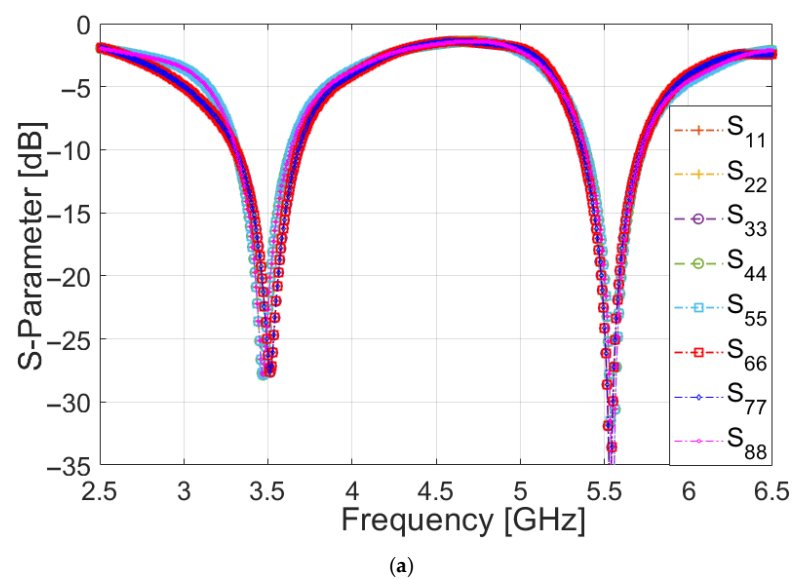


Figure 5. Cont.

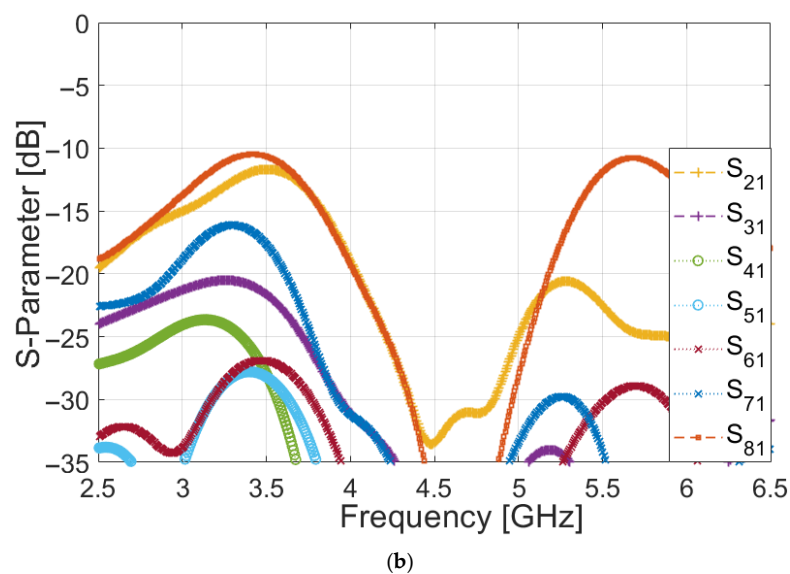


Figure 5. S-parameter results: (a) S_{nn} and (b) S_{n1} of the design.

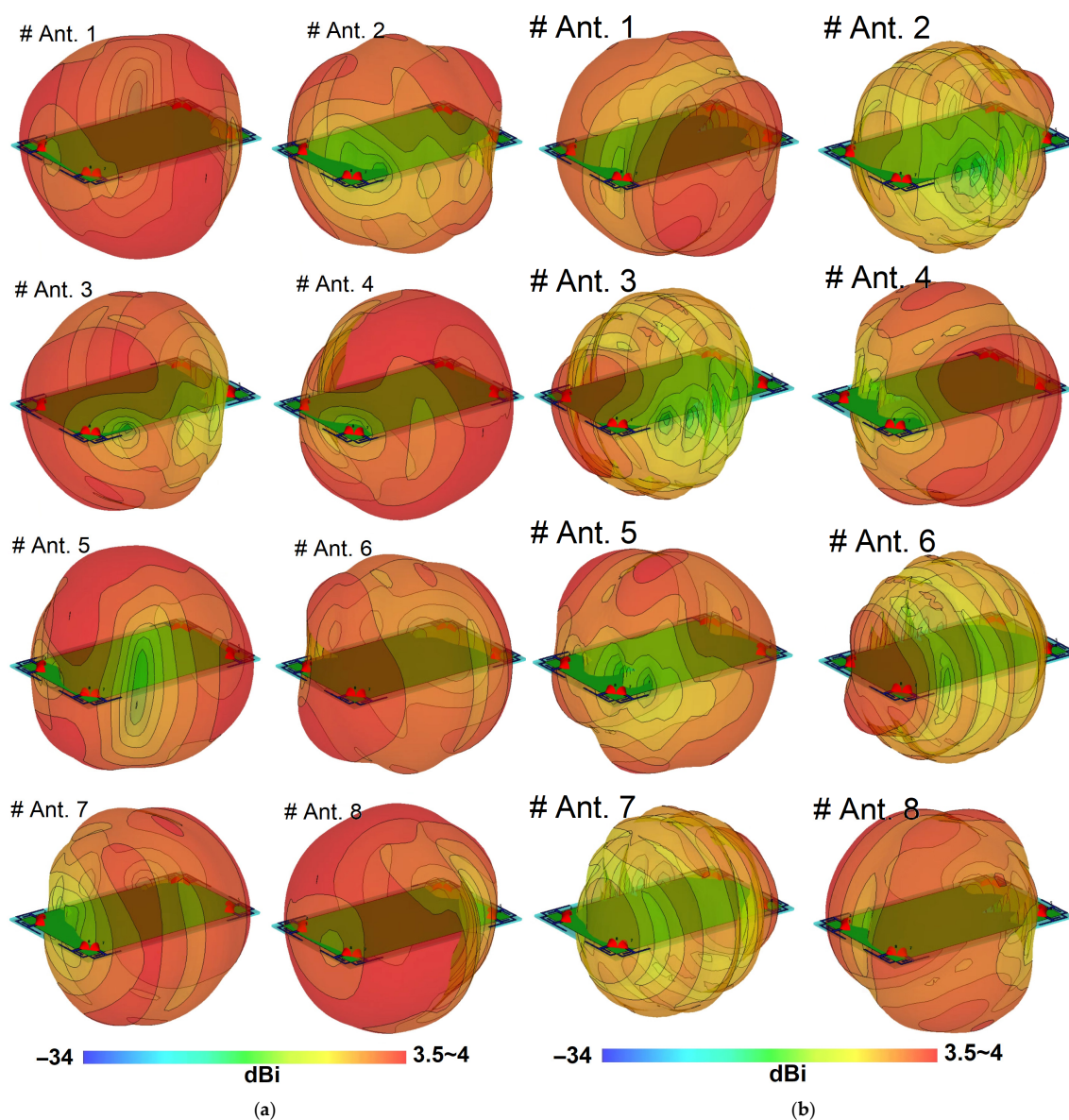


Figure 6. The 3D radiation patterns of the elements at (a) 3.5 GHz and (b) 5.5 GHz.

Figure 7 presents the total efficiency performance of the proposed MIMO antenna system. As shown in the figure, all eight antenna elements exhibit similar efficiency behavior due to the symmetric layout of the design. The total efficiency reaches approximately 70–80% near 3.5 GHz (first band) and about 55–65% around 5.5 GHz (second band), corresponding to the two intended operating bands of the antenna. Outside these bands, the efficiency gradually decreases due to increased impedance mismatch. Slight variations among the antenna elements can be observed, which are mainly attributed to their different positions along the smartphone PCB and the resulting variations in mutual coupling and ground-plane interaction. Nevertheless, the achieved efficiency levels are considered satisfactory for compact MIMO smartphone antennas, where closely spaced elements and limited ground clearance are unavoidable. Overall, the total efficiency results confirm the effective radiation performance of the proposed antenna system in practical handheld-device environments [37].

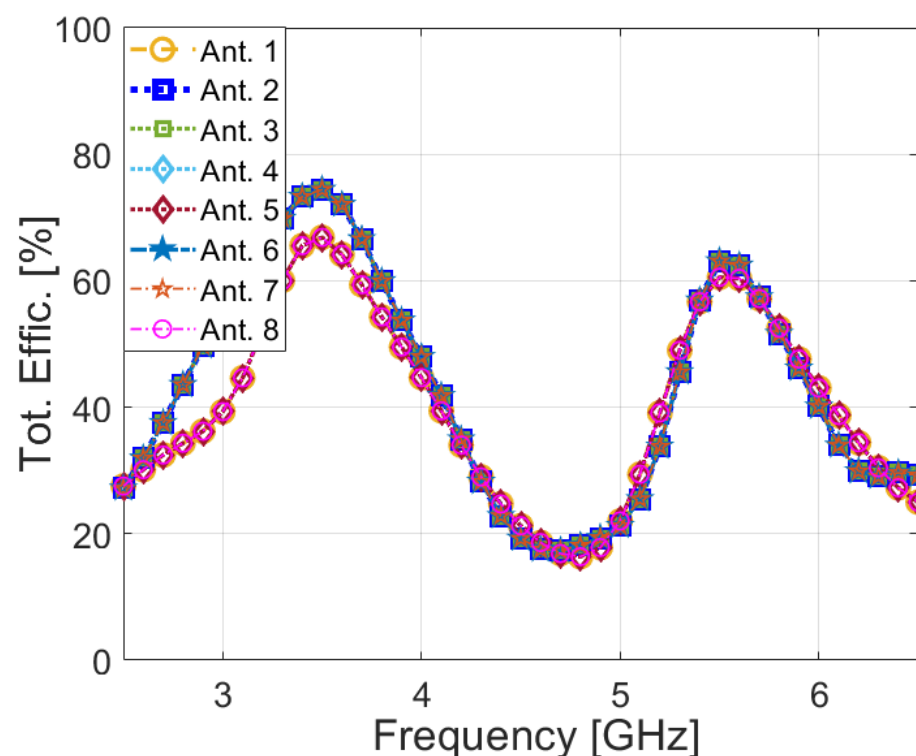


Figure 7. The efficiency results of the dual-band PIFA elements.

3. Prototyping and Measurements

This section describes the fabrication of the proposed MIMO antenna prototype and the corresponding experimental validation. A photograph of the fabricated antenna is shown in Figure 8. The antenna array is realized on a single-layer, low-cost FR4 substrate and consists of an eight-element MIMO configuration, suitable for smartphone integration. For experimental evaluation, the inner conductors of the RF connectors are directly soldered to the antenna elements, while the outer conductors are connected to the ground plane. This feeding arrangement provides reliable excitation of the antenna elements and ensures stable measurement conditions.

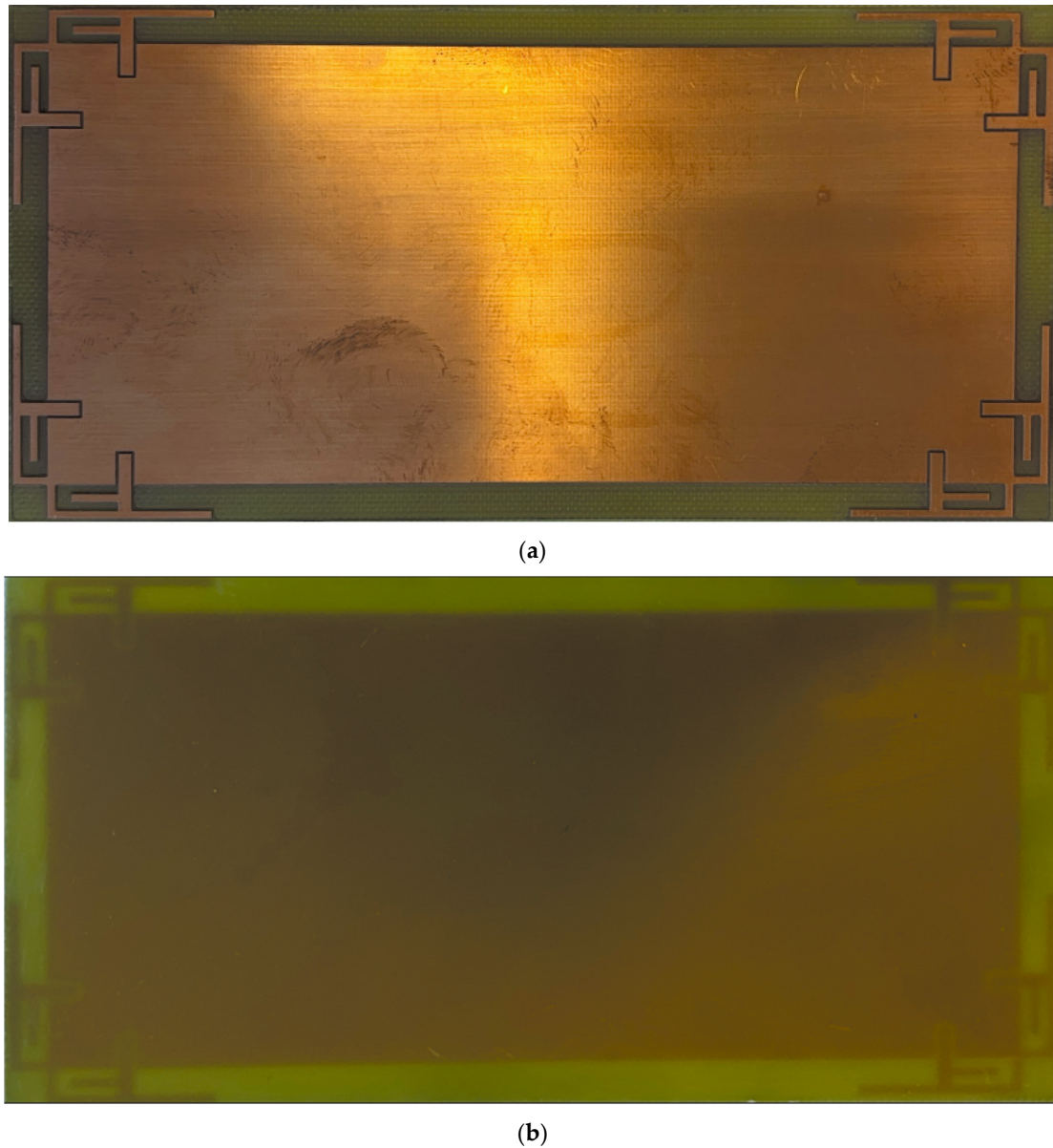


Figure 8. (a) Front and (b) back views of the prototyped sample.

For performance evaluation, the characteristics of ports 1 and 2 are examined and compared, as these ports exhibit similar element placement and radiation behavior. The measured and simulated S-parameter results are presented in Figure 9. As observed, a good agreement is achieved between the measured and simulated S_{11}/S_{22} responses, confirming the validity of the design and the fabrication process. The results also demonstrate that the antenna successfully covers the intended dual-band operating frequencies, validating its suitability for sub-6 GHz 5G applications. The measured results show resonances at 3.5 GHz and 5.5 GHz, with S_{11} below -10 dB across the corresponding operating bands. This dual-bandwidth operation indicates that the antenna is well-suited for a variety of frequency applications. The measured mutual coupling (S_{21}) between adjacent antenna elements remains below -11 dB across the operating frequency bands. This level of isolation confirms that the electromagnetic interaction between the elements is sufficiently suppressed, allowing each antenna element to operate with minimal interference. The measured and simulated S-parameters shown in Figure 9 confirm the expected impedance matching and

isolation characteristics of the antenna. A broader comparison with previously reported smartphone MIMO antennas is summarized in the Comparison section.

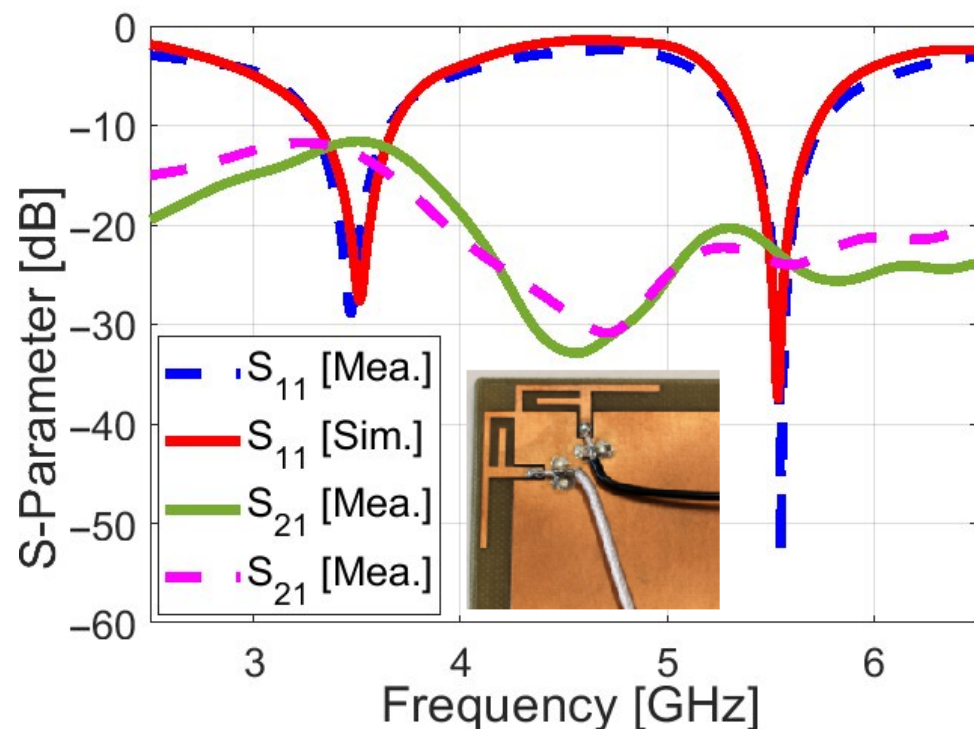


Figure 9. Measured and simulated S-parameters of the antenna pairs.

To further assess the MIMO performance of the proposed antenna system, the envelope correlation coefficient (ECC) and total active reflection coefficient (TARC) are evaluated, as these parameters provide important insight into signal correlation and radiation efficiency in multi-antenna systems [37,38]. Both metrics are calculated using the formulations given below:

$$\text{ECC} = \frac{|S_{mm}^* S_{mn} + S_{nm}^* S_{nn}|^2}{(1 - |S_{mm}|^2 - |S_{mn}|^2)(1 - |S_{nm}|^2 - |S_{nn}|^2)^*} \quad (1)$$

$$\text{TARC} = -\sqrt{\frac{(S_{mm} + S_{mn})^2 + (S_{nm} + S_{nn})^2}{2}} \quad (2)$$

The computed ECC and TARC results, shown in Figure 10a,b, demonstrate favorable MIMO characteristics across the target frequency bands. The ECC remains below 0.005, indicating very low correlation between the antenna elements, which is essential for achieving effective diversity performance. In addition, the TARC values are lower than -20 dB, confirming the efficient radiation of the input power with negligible reflection at the feeding ports. These results collectively verify that the array offers robust MIMO operation with high isolation and excellent diversity. The presented ECC and TARC demonstrate that the antenna array is well-suited for MIMO operation, as it maintains low correlation and optimal radiation efficiency across the target band, providing reliable and effective performance for multi-antenna communication scenarios.

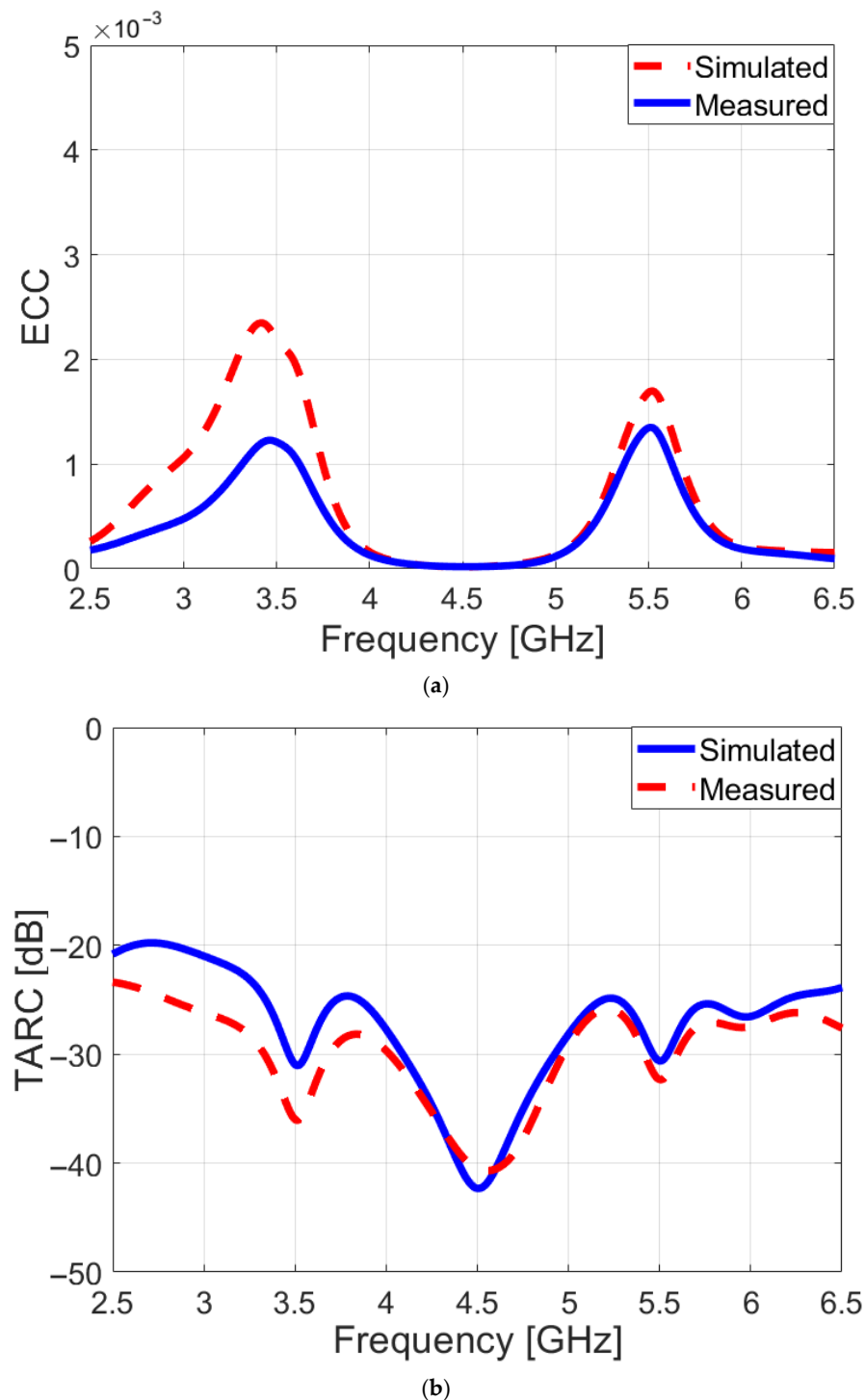


Figure 10. (a) ECC and (b) TARC results.

Figure 11 presents the simulated and measured two-dimensional radiation patterns in the E-plane and H-plane for a representative antenna element at 3.5/5.5 GHz. The radiation patterns were measured inside a fully anechoic chamber using a standard far-field antenna measurement system, where the antenna under test was mounted on a rotational positioner to record the E-plane and H-plane radiation patterns. During the measurements, one antenna port was excited while the remaining port was terminated with a $50\ \Omega$ matched load, ensuring accurate evaluation of the radiation characteristics of an individual element under MIMO operating conditions. At both operating frequencies, the measured radiation patterns show good agreement with the simulated results, validating

the accuracy of the proposed design and fabrication process. The radiation patterns exhibit a quasi-omnidirectional behavior, particularly in the H-plane, which is desirable for smartphone applications to ensure uniform signal coverage regardless of device orientation. In the E-plane, the patterns display slight nulls and shape variations, which are primarily attributed to the antenna geometry, ground-plane effects, and the proximity of adjacent antenna elements. At 3.5 GHz, the radiation patterns are relatively smooth and stable, indicating that the fundamental resonant mode dominates radiation. At 5.5 GHz, increased pattern variations and minor distortions are observed, which are expected due to the excitation of higher-order modes and the more complex current distribution associated with the added strip. Nevertheless, the overall radiation remains well distributed, with no deep nulls that would severely degrade coverage. The observed quasi-omnidirectional radiation behavior confirms that the proposed array is well-suited for practical scenarios, where the device may be held in various orientations and positions. This radiation stability, combined with the demonstrated multi-band and MIMO capabilities, ensures reliable performance requiring consistent coverage [39,40].

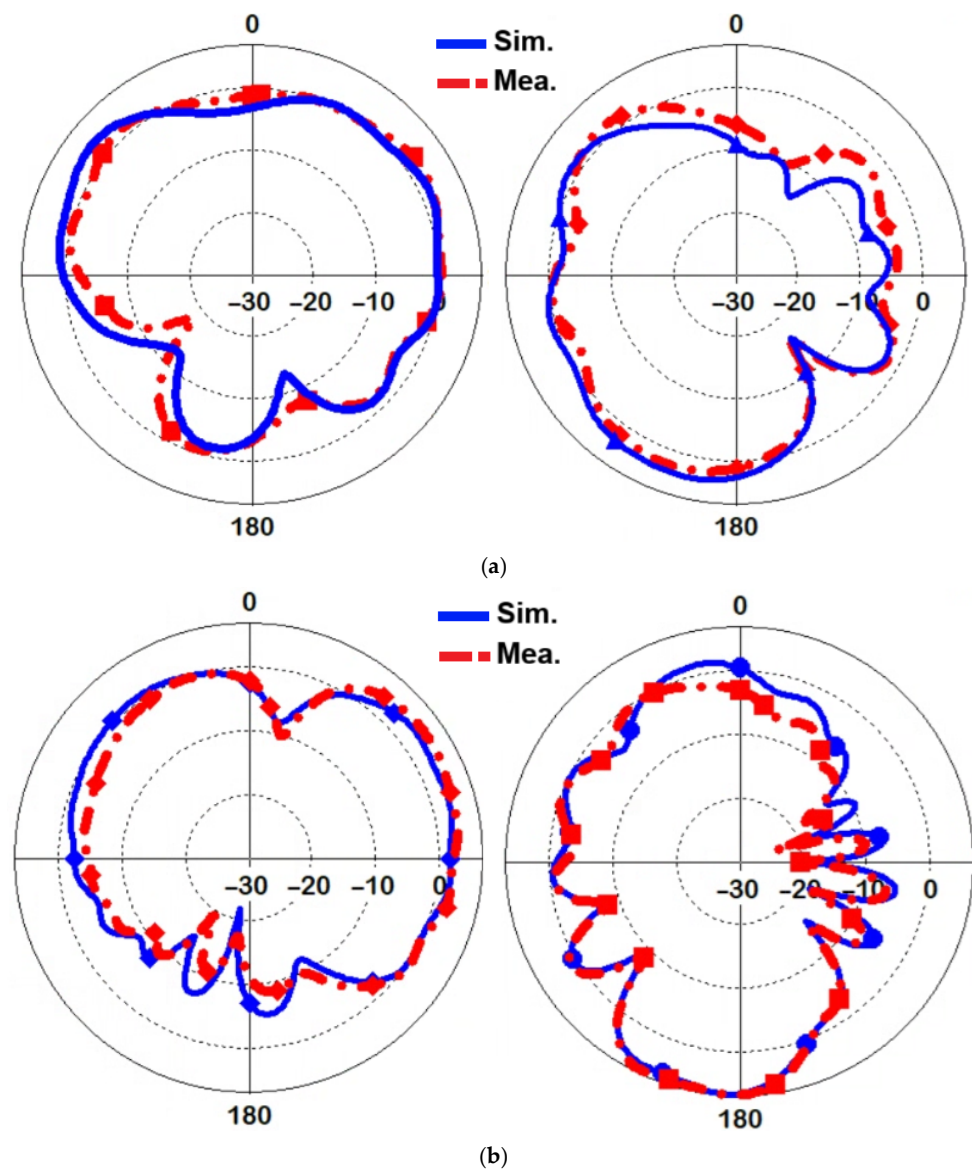


Figure 11. The 2D radiation patterns (E/H-planes) of the element at (a) 3.5 GHz and (b) 5.5 GHz.

4. SAR User-Impact Analyses

The specific absorption rate (SAR) is used to evaluate the amount of RF energy deposited in human tissue and is expressed in W/kg. SAR assessment is an essential safety requirement for mobile communication devices, as it verifies compliance with international exposure guidelines and limits potential biological effects [41,42]. The SAR magnitude is influenced by several factors, including antenna configuration, operating frequency, device placement, and the relative position of the user. To ensure user safety, regulatory organizations such as the Federal Communications Commission (FCC) and the International Commission on Non-Ionizing Radiation Protection (ICNIRP) have defined maximum allowable SAR thresholds, commonly set at 2 W/kg, to avoid excessive tissue heating [43]. The SAR performance of the proposed MIMO antenna array under talk-mode conditions is illustrated in Figure 12. The simulation setup is shown in Figure 12a, while the corresponding SAR distributions at 3.5 GHz and 5.5 GHz are presented in Figure 12b,c, respectively. The results indicate noticeable variation in SAR levels across the antenna elements due to their different positions relative to the head and hand phantom. In particular, Antenna 1 exhibits the highest SAR, whereas Antenna 4 produces the lowest SAR value during talk-mode operation. This variation in SAR values is primarily attributed to the relative placement of the antenna elements with respect to the head and hand phantom. Antenna 1 is located closer to the phantom, leading to stronger electromagnetic coupling and higher localized energy absorption. In contrast, Antenna 4 is positioned farther from the phantom, resulting in reduced coupling and lower SAR levels. These observations confirm that increasing the separation distance between the radiating elements and the user's body effectively reduces SAR, while all SAR values remain within the safety limits, demonstrating the suitability of the design for practical applications.

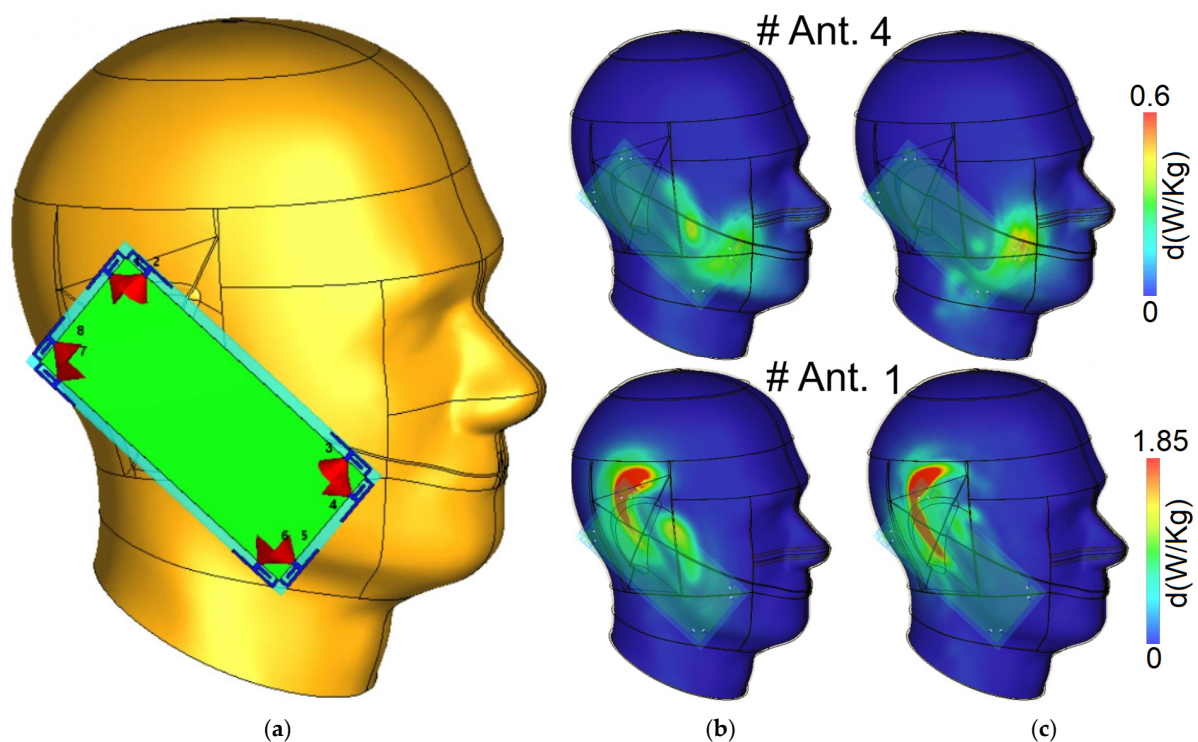


Figure 12. (a) Simulation setup and SAR distributions at (b) 3.5 and (c) 5.5 GHz.

Figure 13 illustrates the three-dimensional radiation patterns of the antenna elements at the resonance frequencies of 3.5/5.5 GHz when the device is held by both hands. As expected, user-hand interaction introduces absorption and scattering effects that lead to some

degradation in radiation performance. Nevertheless, the proposed antenna array retains adequate gain levels and broad spatial coverage across all antenna elements. Although pattern distortions and localized gain reductions are observed for elements positioned closer to the hands, no severe radiation nulls appear. The variation in radiation behavior among the elements is mainly attributed to their different locations on the smartphone PCB and their proximity to the user's hands. Overall, the results demonstrate that the proposed antenna design maintains robust radiation performance under realistic hand-holding conditions, confirming its suitability for practical applications [44,45].

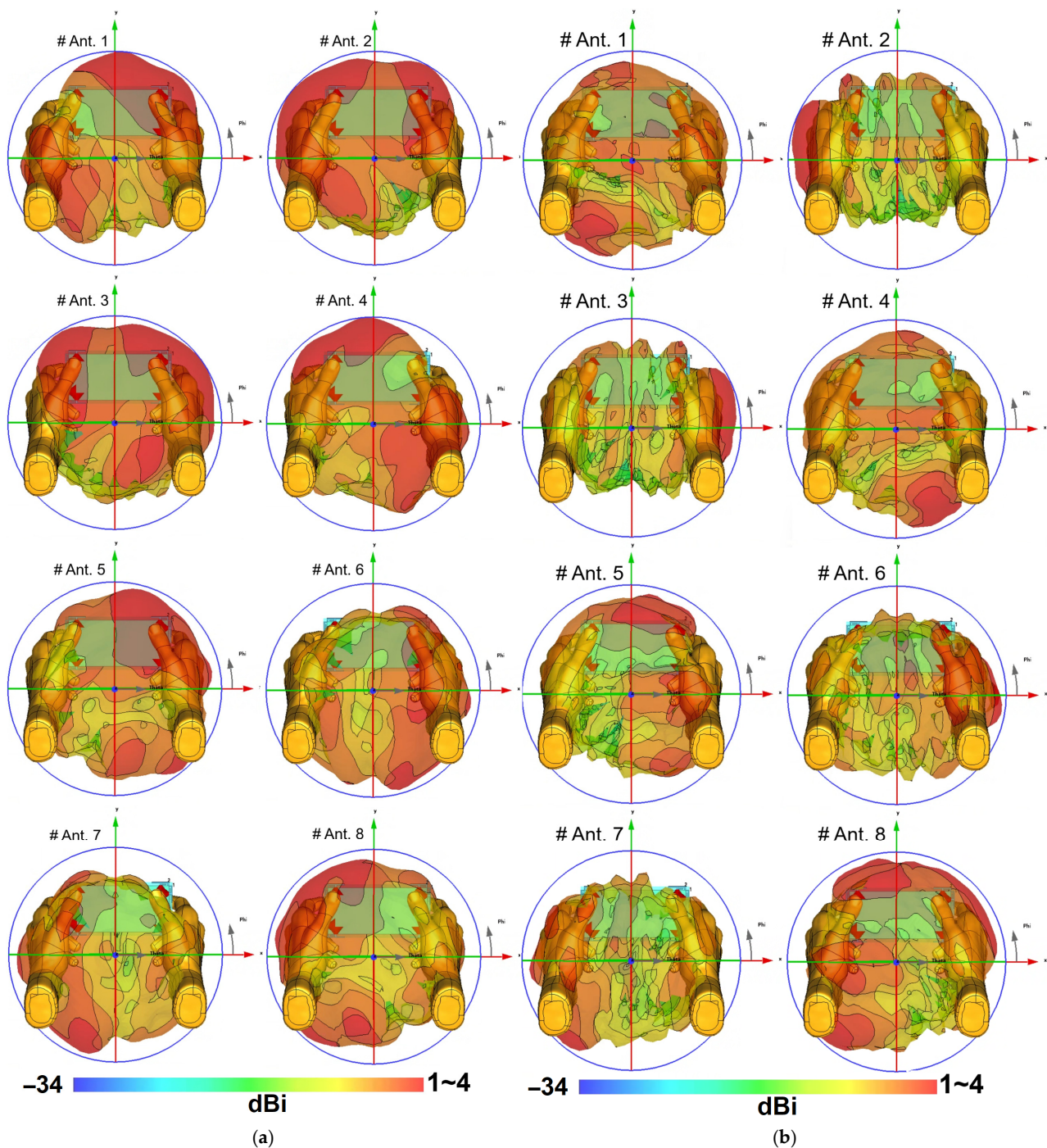


Figure 13. The 3D radiation patterns of the elements at (a) 3.5 GHz and (b) 5.5 GHz.

5. Comparison

A comparative evaluation of the introduced antenna and several existing designs from the literature is presented in Table 2. Key performance metrics, such as operating bandwidth, efficiency, isolation, SAR, and ECC, are compared to highlight the advantages of the introduced antenna. The comparison shows that many existing antennas operate over limited frequency ranges or require complex structures to achieve acceptable MIMO performance. In contrast, the proposed array exhibits a multi-resonant characteristic that enables dual-band operation covering 3.3–3.7 GHz and 5.3–5.8 GHz, addressing critical sub-6 GHz 5G bands. Furthermore, the proposed design employs a planar, dual-polarized configuration, which enhances polarization diversity while maintaining a compact form factor suitable for smartphone platforms. An additional advantage of the proposed antenna is its fully planar structure with ease of single-layer PCB integration, which significantly simplifies fabrication and prototyping compared to multilayer or three-dimensional designs. Despite the closely spaced antenna elements, the design achieves competitive efficiency, adequate isolation without decoupling structures, and an exceptionally low ECC, indicating excellent diversity performance.

Table 2. Performance comparison between the proposed and reported designs.

Ref.	Year	Bandwidth (GHz)	Efficiency (%)	Size (mm ²)	Isolation (dB)	ECC	Diversity	Planar	Advantages	Limitations	SAR
[10]	2025	3.4–3.6	70–80	160 × 75	−18	N/A	No	Yes	High efficiency Good isolation	Four element High SAR Single band	2.6
[11]	2019	3.4–3.6	50–55	136 × 68	15	<0.10	No	No	Self-isolated structure	Low efficiency Non-planar Narrow band	N/A
[12]	2021	3.4–3.6	60–75	150 × 80	17	<0.05	No	Yes	Simple design	Large element Narrow band	N/A
[13]	2017	3.4–3.6	60–70	150 × 75	11	<0.15	No	No	Gap coupled Compact element	Non-planar Single band	N/A
[14]	2025	3.4–3.6	55–70	150 × 75	12	<0.10	No	No	Planar structure Narrow band	Complex element High ECC	N/A
[15]	2021	3.45–3.55	50–68	130 × 50	15	<0.30	No	No	Good SAR	Non-planar Narrow band	1.6
[16]	2022	3.4–3.6	50–78	150 × 77	10	< 0.1	No	Yes	High isolation	Single band No diversity	N/A
[17]	2022	3.45–3.55	50–70	154 × 74	15	<0.1	No	No	Compact element	High SAR Non-planar Narrow band	2.3
[18]	2023	3.3–3.8	40–75	150 × 75	13	<0.06	No	Yes	Planar element Tunable	Complex element Single band	N/A
[19]	2023	4.4–5	40–80	150 × 75	10	<0.1	No	Yes	Wide-band Planar element	Large elements Single band	1.96
[20]	2023	3.4–3.6	40–75	150 × 75	19	<0.1	Yes	Yes	Diversity Planar element	Large elements Single band	1.8
[21]	2022	3.5–3.7	50–80	150 × 75	16	<0.03	No	No	Good efficiency	Non-planar Single band	N/A
[22]	2024	3.3–3.5 4.6–4.8	65–70	150 × 80	18	<0.1	Partially	Yes	Dual band	Partial diversity No SAR analysis	N/A
[23]	2024	3.75–3.85 5.35–5.45	40–85	150 × 74	13	<0.01	No	Yes	Planar structure Good SAR	Two element Narrow band	1.2
[24]	2021	3.4–3.6 4.8–5	60–80	150 × 75	10	<0.1	No	No	Dual band	Four element Non-planar	N/A
[25]	2022	3.4–3.6 5.15–5.925	60–85	150 × 75	10	<0.1	No	No	Good ECC Dual band	Large elements Simulations only	N/A
[26]	2023	3.4–3.6 5.2–5.7	60–65	150 × 75	10	<0.01	No	No	Dual band	Large elements No diversity	N/A
This Work	2026	3.3–3.7 5.3–5.8	55–80	150 × 75	11	<0.002	Yes	Yes	Dual band dual polarization planar single-layer Low ECC	Moderate isolation	1.8

Moreover, SAR analysis confirms low SAR levels and good user interaction, ensuring compliance with international safety standards under realistic usage scenarios. Overall, the comparison demonstrates that the proposed MIMO antenna offers a good combination of wideband performance, dual-polarization, structural simplicity, low SAR, and practical implementability. From Table 2, it can be observed that the proposed antenna distinguishes itself by simultaneously providing dual-band operation (3.3–3.7 GHz and 5.3–5.8 GHz), dual polarization, planar single-layer implementation, and very low ECC (<0.002), while maintaining acceptable isolation without requiring dedicated decoupling structures. These combined features represent the main contribution of this work and make the antenna suitable for integration in compact handheld wireless devices.

6. Conclusions

A compact dual-band/dual-polarized MIMO antenna array suitable for sub-6 GHz 5G smartphone applications has been presented. The design employs a planar PIFA-based configuration integrated on a $150 \times 75 \text{ mm}^2$ smartphone PCB, enabling simple fabrication and ease of prototyping using standard PCB processes. By incorporating an additional strip into the antenna structure, dual-band operation at 3.3–3.7 GHz and 5.3–5.8 GHz is achieved, covering key sub-6 GHz 5G bands. The antenna array achieves efficiency values between approximately 55% and 80% while maintaining isolation of about 11 dB between closely spaced elements without requiring dedicated decoupling structures. In addition, the design exhibits a very low envelope correlation coefficient ($\text{ECC} < 0.002$) and TARC below -20 dB , indicating excellent diversity behavior and confirming effective MIMO and polarization diversity operation. Satisfactory efficiency and stable radiation patterns are maintained in both free-space and user-hand scenarios, indicating robustness under realistic operating conditions. SAR analysis also confirms compliance with international safety standards, with a maximum value of approximately 1.8 W/kg under typical operating conditions. Overall, the suggested antenna offers a good combination of wideband performance, dual polarization, planar structure, low SAR, and practical implementability, making it a suitable candidate for current 5G smartphones and future multi-band sub-6 GHz handheld devices evolving toward 6G wireless systems.

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